paper

2026-03-18

==========

INTRODUCTION

==========

Online film ratings play a crucial role in shaping consumer viewing decisions and form a key input for recommendation systems on streaming platforms. As a result, understanding which characteristics of audiovisual content are associated with higher audience ratings is of both theoretical and practical importance. One potentially relevant characteristic in this context is runtime.

Previous research suggests that runtime may influence audience evaluations. For example, Moon et al. (2009) find that longer films tend to receive higher ratings on average. One explanation is that longer films may be perceived as offering greater narrative depth, complexity, and overall value, which can positively affect audience evaluations. In this view, runtime functions as a signal of quality.

At the same time, alternative theoretical perspectives suggest the opposite effect. According to Attention Restoration Theory, prolonged exposure to audiovisual stimuli may lead to cognitive fatigue and reduced attentional capacity (Baumgartner & Kühne, 2024). This decrease in attention may negatively affect viewers' ability to process and appreciate a film, potentially resulting in lower evaluations. Taken together, these perspectives lead to contrasting theoretical expectations: while longer films may enhance perceived quality, they may simultaneously reduce viewer engagement due to cognitive fatigue. As a result, theoretical perspectives provide conflicting expectations regarding the relationship between runtime and audience ratings.

Moreover, the effect of runtime is unlikely to be uniform across all types of films. Different genres create distinct expectations regarding narrative pacing, story development, and appropriate length. For example, longer runtimes may be more acceptable or even expected in genres such as drama or biography, whereas shorter and faster-paced formats may be more suitable for genres such as comedy or horror. This suggests that genre may play an important moderating role in the relationship between runtime and ratings.

Based on these considerations, the central research question is formulated as follows:

To what extent does runtime predict the average IMDb rating of films, and to what extent is this relationship moderated by genre?

This research question aligns well with the available IMDb datasets and focuses on whether movie length is associated with perceived quality, as reflected in average IMDb ratings. In addition, the number of votes may affect the reliability and stability of ratings. Movies with only a small number of votes may receive more extreme or biased evaluations. Therefore, the number of votes is included as a control variable. Release year is also included as a control variable to account for potential temporal trends in IMDb ratings. Audience preferences, rating behaviour, and industry standards may have changed over time, which could systematically influence how films are evaluated. For example, more recent films may receive different ratings due to changing expectations, technological advancements, or shifts in genre popularity. By controlling for release year, the analysis accounts for these time-related effects and provides a more accurate estimate of the relationship between runtime and average rating.

The findings of this study are relevant for both consumers and platforms. For consumers, they provide insight into how to interpret online ratings in relation to film characteristics such as runtime. For platforms, the results may inform the design of recommendation algorithms that rely on rating data, by clarifying whether and how runtime contributes to perceived content quality.

==========

DATA

==========

The data used in this study are obtained from the official IMDb datasets, specifically we used these two datasets:

- title.basics.tsv.gz
- title.ratings.tsv.gz

These datasets are programmatically downloaded within the R script to ensure full reproducibility. The title.basics dataset contains **12,373,014** observations and includes information on title characteristics such as title type, runtime, and genre. The title.ratings dataset contains **1,650,357** observations and provides the average user rating and the number of user votes. The two datasets are merged using the unique identifier tconst.

To ensure a consistent and comparable sample, the analysis is restricted to movies (titleType = "movie"). Titles with missing values for runtime, average rating, or number of votes are excluded.

Because IMDb assigns multiple genres to some titles, only the first listed genre is retained as the primary genre classification to avoid duplicate observations and ensure one observation per film. After cleaning and filtering, the final merged dataset contains **299,473** unique movie observations.

==========

VARIABLES

==========

Dependent Variable: Average Rating

The dependent variable is the average IMDb user rating of a movie, measured by `averageRating`. It is recorded on a continuous scale from 1 to 10 and reflects audience evaluations of the content. Higher values indicate more favourable audience evaluations, while lower values reflect more negative evaluations.

Independent Variable: Runtime

The independent variable is the runtime of a movie, measured by `runtimeMinutes`. Runtime is expressed in minutes and treated as a continuous variable. It represents the total length of the content. To ensure a consistent and comparable sample, extreme runtime values are excluded from the analysis. Movies shorter than 40 minutes and longer than 240 minutes are removed, as these are unlikely to represent standard feature films.

Control Variable: Number of Votes

The control variable is the total number of IMDb user votes, measured by `numVotes`. This variable captures the popularity and visibility of a title. Because the distribution of vote counts is strongly skewed, the variable may be log-transformed in the regression analysis. This log-transformation results in a more normally distributed variable suitable for linear regression.

Control Variable: Release Year

Release year of the movie is included as an additional control variable, measured by `startYear`. This variable captures the year in which the movie was released. The control variable is treated as a continuous variable and it accounts for potential temporal trends in iMDb ratings, as it may be that audience evaluation standards and rating behaviour may have changed over time.

Moderator: Genre

Genre is included as a moderating variable to test whether the relationship between runtime and rating differs across movie types. Since IMDb allows multiple genres per title, only the first listed genre is used as the primary genre classification in order to preserve one observation per film.

Genre is treated as a categorical variable and operationalised using dummy variables in the regression analysis, with one genre serving as the reference category. For interpretability, genres are grouped into the ten most frequent categories, with all remaining genres combined into an `"Other"` category.

==========

METHOD

==========

To answer the research question, linear regression analysis is employed, as both the dependent variable (average IMDb rating) and the independent variable (runtime in minutes) are continuous. Prior to analysis, the data were prepared through several cleaning steps. The two IMDb datasets (title.basics and title.ratings) were merged using the unique identifier tconst and filtered to retain only feature films (titleType = "movie"). Runtime outliers were removed by restricting observations to films between 40 and 240 minutes, based on inspection of the empirical quantile distribution. Titles with missing values on runtime, average rating, number of votes, or release year were excluded via listwise deletion, yielding a final analytic sample of 299,473 films.

To address the skewed distribution of the number of votes, this variable was log-transformed (log_votes = log(numVotes + 1)) before inclusion as a control variable, improving the normality of its distribution and reducing the influence of highly voted outliers. Because IMDb assigns multiple genres per title, only the first-listed genre was retained as the primary genre classification (main_genre) to maintain one observation per film. Genres were subsequently grouped into the ten most frequent categories, with all remaining genres collapsed into an "Other" category, resulting in an eleven-level categorical moderator (genre10).

The moderation hypothesis is tested by including interaction terms between runtime and genre in the regression model, allowing the slope of runtime to vary across genre categories. Release year was additionally examined as a potential covariate through correlation analysis. All analyses were conducted in R (version 4.0 or higher).

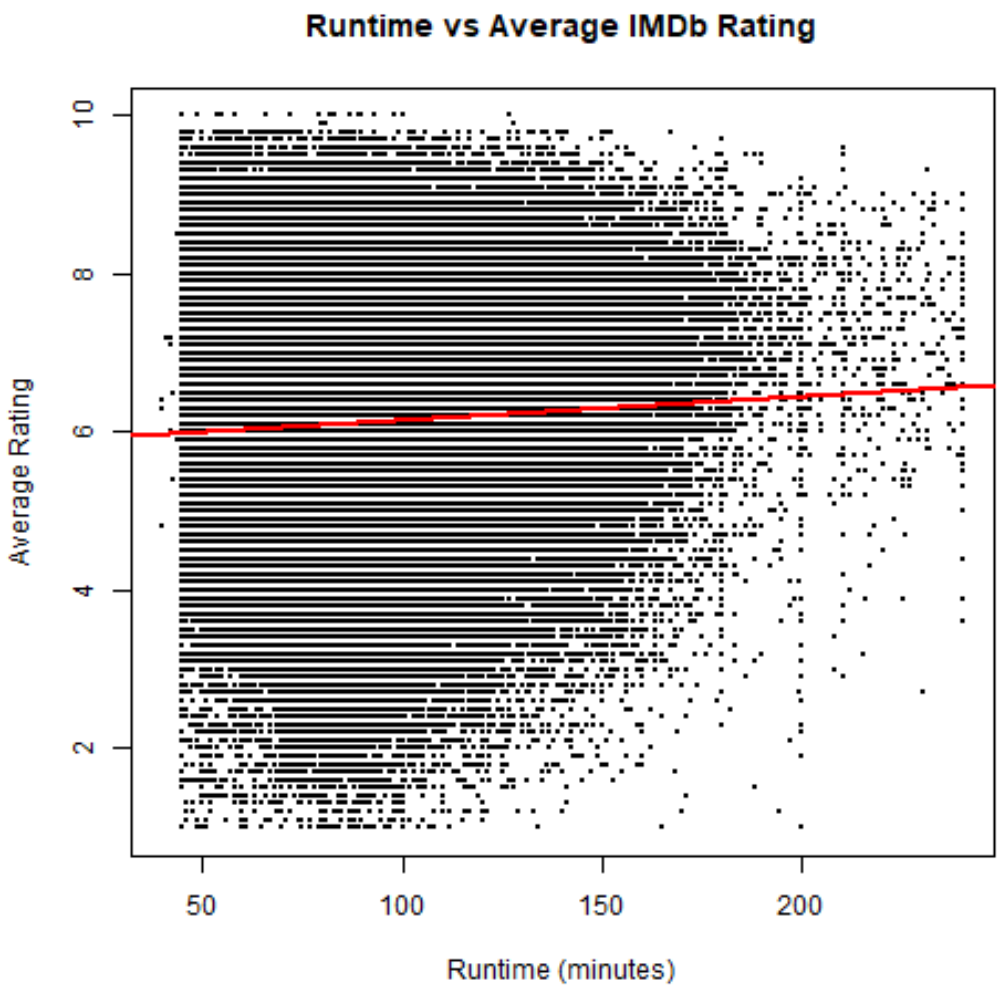
==========

PREVIEW OF FINDINGS

==========

Preliminary analysis shows that the relationship between runtime and average IMDb rating is statistically very weak. The Pearson correlation between runtime and average rating is approximately r = 0.008, which is very close to zero and indicates that longer movies do not systematically receive higher ratings. This challenges the assumption that longer films tend to be rated more favourably (Moon et al., 2009), suggesting that runtime is not a reliable determinant of perceived quality.

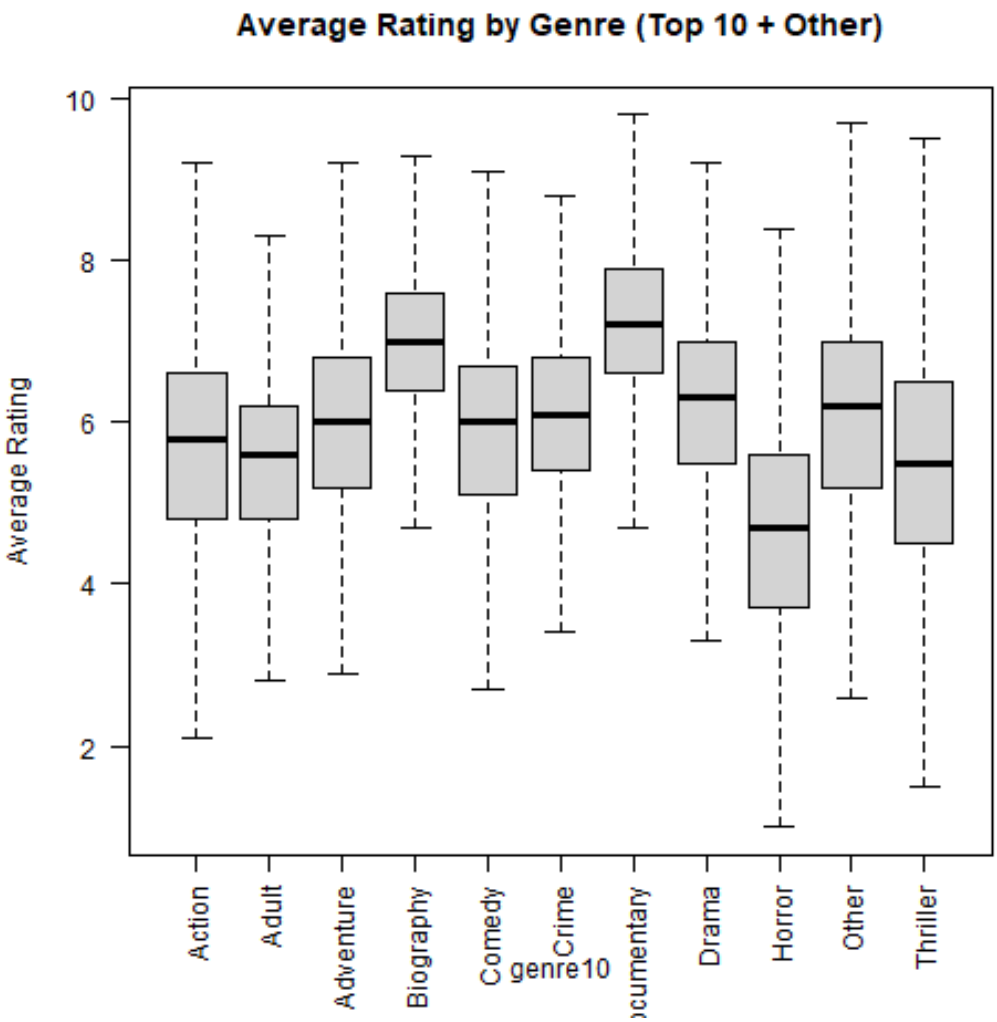
This pattern is also visible in the scatterplot below, where the regression line shows only a slight positive trend and the data points are widely dispersed. This suggests substantial unexplained variation in ratings across movies with similar runtimes. In other words, runtime alone does not appear to be a strong predictor of the average IMDb rating of a movie.



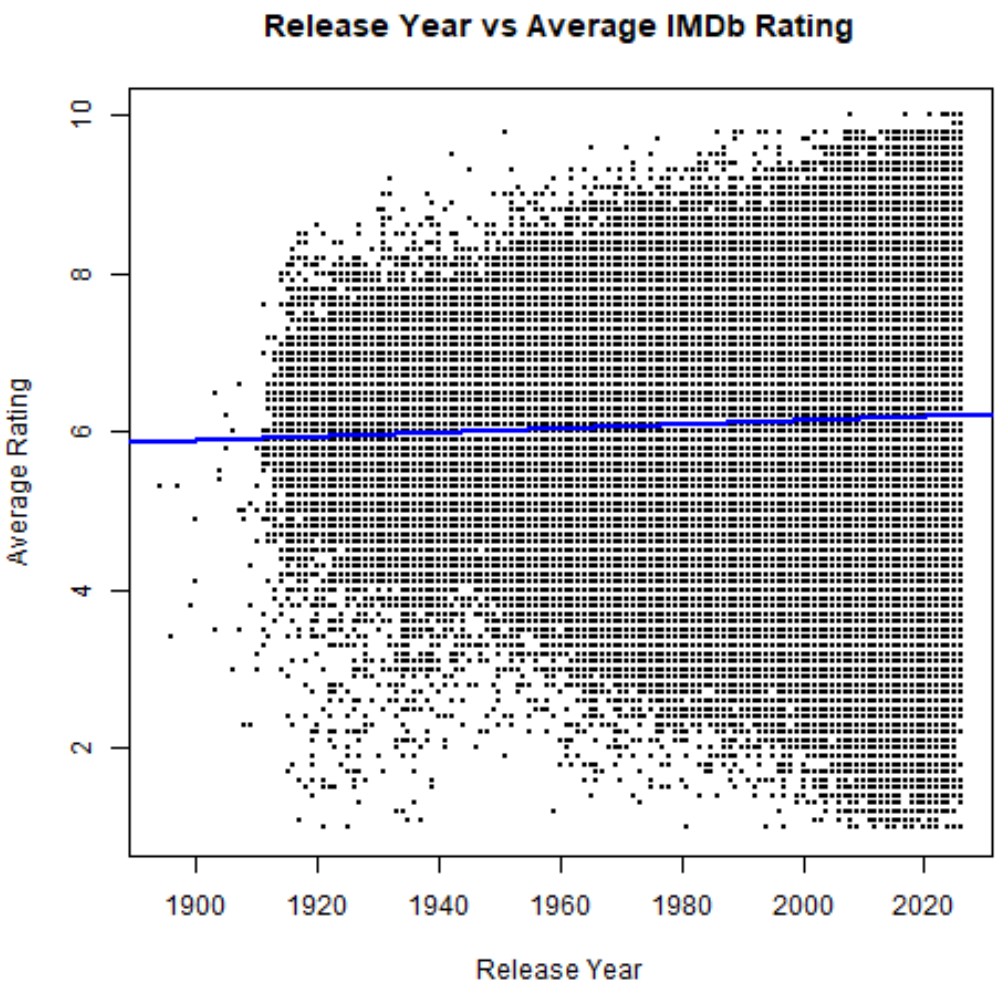
In other words, runtime alone does not appear to be a strong predictor of the average IMDb rating of a movie.

The correlation between the log-transformed number of votes and average rating is approximately **-0.08**, indicating a very small negative relationship. This may reflect a diversification effect, whereby wider viewership introduces more heterogeneous and critical evaluations. Including the number of votes as a control variable is therefore analytically important, as it isolates the effect of runtime from variation in title visibility, thereby strengthening the internal validity of the estimates.

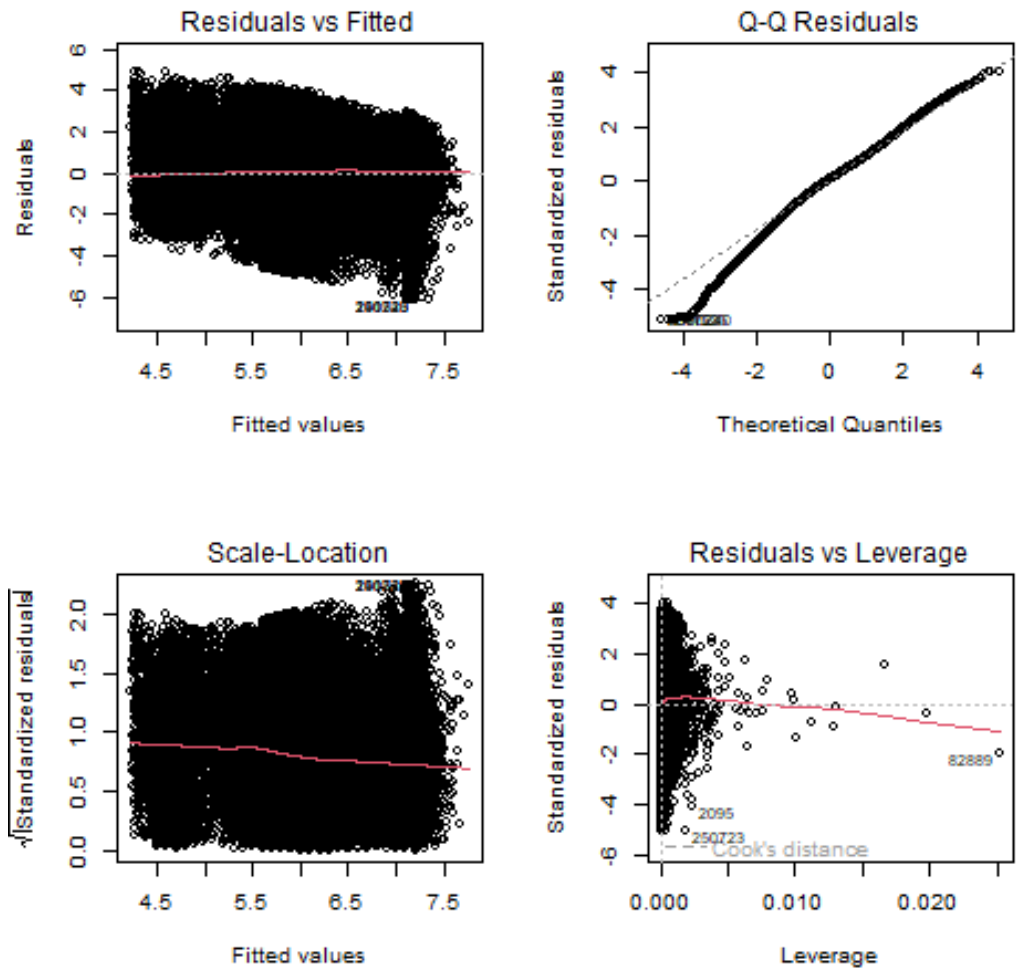
As shown in the boxplot below. Average ratings vary substantially across genres. For example, the average ratings of the genres *Biography* and *Documentary* are noticeably higher than those of *Horror*. This suggests that genre may play an important moderating role in the relationship between runtime and rating.



Further analysis shows trends over time:



Finally, we assess the validity of the regression model:



==========

REPRODUCIBILITY

==========

The findings of this study are presented in a structured R Markdown report. In this report, the complete data analysis is documented step by step. The report includes data preparation, descriptive statistics, regression analysis, and visualisations such as scatterplots and boxplots.

Furthermore, the workflow is fully reproducible. The IMDb data can be downloaded automatically and analysed through the script. This allows other researchers to replicate the study easily or extend the analysis using different selection criteria, such as focusing on a specific genre. All scripts are version-controlled using Git and hosted in a public GitHub repository, ensuring full transparency and accessibility of the analytical workflow.

==========

INTERPRETATION AND RELEVANCE

==========

The findings suggest that runtime, on its own, is a very weak predictor of the average IMDb rating. This is relevant because it challenges the assumption that longer movies necessarily receive higher ratings (Moon et al., 2009). In practical terms, this indicates that longer runtime is not a reliable strategy for achieving higher ratings.

In addition, the results show substantial variation in ratings across genre categories. This implies that general statements about an "ideal movie length" may be misleading and that such conclusions should instead be approached in a genre-specific manner.

Including the number of votes as a control variable is essential for a more accurate interpretation of the results. By controlling for the number of votes, the effect of runtime is estimated while accounting for variation in title visibility and popularity, which strengthens the internal validity of the analysis. Including the release year as an additional control variable further strenghtens the analysis by accounting for potential changes in rating behaviour over time. Together, these two control variables help ensure that the estimated effect of runtime on average rating is not confounded by differences in movie popularity or the time period in which a film was released.

==========

REPOSITORY OVERVIEW

==========

The repository is organised to ensure reproducibility and transparency. Scripts for data acquisition, preparation, and analysis are separated into dedicated subdirectories, and the master makefile automates the complete pipeline from data download to final output.

Repository overview

- `data/`
  - `title.basics.tsv.gz`
  - `title.ratings.tsv.gz`
- `reporting/`
  - `final_report.Rmd`
- `src/`
  - `analysis/`
    - `analysis.R`
    - `makefile`
  - `data-preparation/`
    - `data-cleaning.R`

```
        * download-data.R
        * makefile
```

- `README.md`
- `makefile`

==========

DEPENDENCIES

==========

This project was developed using R (version 4.2.0 or higher) and RStudio. The following R packages are required to run the workflow:

- `readr` for importing the IMDb .tsv.gz datasets
- `dplyr` for data filtering, merging, and variable construction
- `tidyr` for data reshaping and handling missing values
- `rmarkdown` for rendering the R Markdown reports
- `knitr` for document knitting and output generation

If any of these packages are not yet installed, they can be installed collectively using:

```
install.packages(c("readr", "dplyr", "tidyr", "rmarkdown", "knitr"))
```

==========

RUNNING INSTRUCTIONS

==========

To reproduce the results of this workflow, follow these steps:

Step 1: Clone the repository

Using Command Prompt (Windows) or Terminal (Mac), clone the repository to your local machine:

```
git clone https://github.com/course-dprep/The-effect-of-movie-runtime-on-average-IMDb-ratings.git
```

Once cloned, all necessary folders and scripts will be available locally.

Step 2: Download the IMDb datasets

Navigate to src/data-preparation/ and run download-data.R. This script automatically creates a data/ folder and downloads the required IMDb datasets:

```
# Create data folder
dir.create("data", showWarnings = FALSE, recursive = TRUE)

# Download IMDb datasets
download.file("https://datasets.imdbws.com/title.basics.tsv.gz",
              "data/title.basics.tsv.gz")

download.file("https://datasets.imdbws.com/title.ratings.tsv.gz",
              "data/title.ratings.tsv.gz")
```

Step 3: Load packages and read data

Run data-cleaning.R to load the required packages and import the downloaded datasets into RStudio:

```
# Load required libraries
library(readr)
library(dplyr)
library(tidyr)

# Read datasets
basics  <- read_tsv("data/title.basics.tsv.gz", na = "\\N")
ratings <- read_tsv("data/title.ratings.tsv.gz", na = "\\N")
```

After completing these steps, the full analysis pipeline, including data merging, cleaning, variable construction, and regression analysis can be reproduced by knitting final_report.Rmd in RStudio or by executing the master makefile from the terminal. No manual data preprocessing is required.

==========

ABOUT

==========

This project is set up as part of the Master's course Data Preparation & Workflow Management at the Department of Marketing, Tilburg University, the Netherlands.

The project is implemented by team 1 members:

- **Bente van Brussel**: b.j.m.vanbrussel@tilburguniversity.edu
- **Niels Deenen**: n.deenen@tilburguniversity.edu
- **David Lindwer**: d.j.lindwer@tilburguniversity.edu
- **Demi Verburg**: d.verburg@tilburguniversity.edu
- **Marijn van Dooren**: m.vandooren@tilburguniversity.edu